

1 Turing Machine for Palindromes

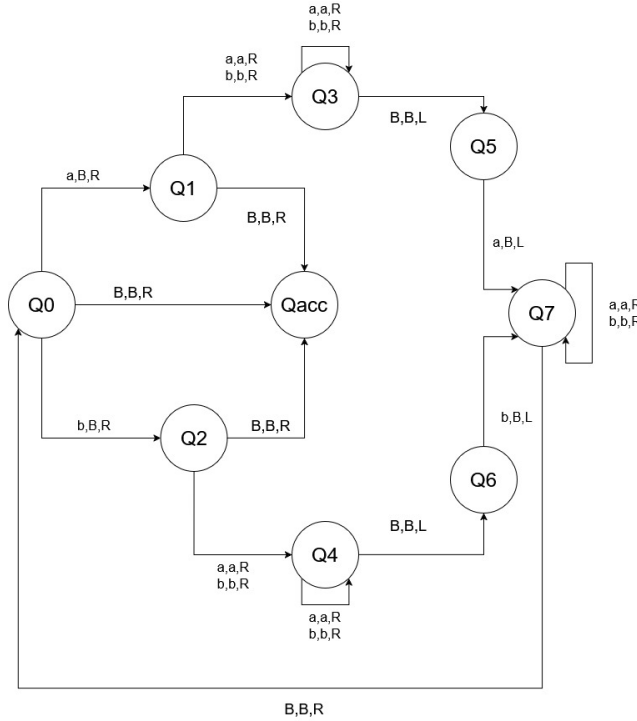
Question: Construct a Turing Machine M that recognizes the language of all palindromes over the alphabet $\Sigma = \{0, 1\}$.

For an alphabet $\Sigma = \{a, b\}$, we have to construct a Turing Machine M that recognizes all palindromes over the alphabet. The core idea for the solution for this task will be an algorithm as follows:

1. Read the first character of the string and remember it.
2. Traverse to the last character and see if it matches the first alphabet of the string.
3. If the two symbols match then the machine performs step 1 and 2 with the next pair of the alphabets until all pairs $i, n - i$ for $i = \lfloor n/2 \rfloor$ or for odd length palindromes only the $n/2$ symbol remains. (n is the length of the string).

The smallest length palindrome will be ϵ for even length palindromes and a/b for odd length palindromes.

The final state diagram that follows the above steps will look like this:



- Here Q0 represents the start state and Qacc is the final state.
- Q1, Q2 perform the step of remembering the first character of the string by representing two different states for the inputs.
- Q3, Q4 Traverse to the end of the string and their transition to Q5,Q6 simply helps us trace back to the last step of the string.
- Q5, Q6 checks whether the character at the end matches the the character that transitioned to Q1, Q2 respectively.
- Q7 then traverses back to the start of the alphabet and then finally ggoes to Q0 to repeat the process once it reaches the first character again.
- The transitions (Q0, Q1), (Q0, Q2), (Q5, Q7), (Q6,Q7) all replace their input character with B so that we can repeat the process for all pairs as once the first and last character are confirmed to be the same, we do not need to verify them ever again and this helps in maintaining the loop.
- The transitions (Q0, Qacc), (Q1, Qacc), (Q2, Qacc) represent the acceptance of the shortest palindromic strings. Since we replace our characters with B eventually all valid palindromes will transform into the form of their shortest palindrome depending on if they are even or not.